

# W02

MATH 135

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# 1 Q1

## 1.1 (a)

**Problem 1.** Let  $K$ ,  $L$ , and  $M$  be statement variables and suppose that the following logical expression is TRUE:

$$\neg((L \vee (\neg M)) \implies (\neg K)) \wedge (\neg L).$$

Determine with justification all possible truth value combinations of  $K$ ,  $L$ , and  $M$ .

**Solution.** According to the extra lecture, we have already proved that

$$(A \implies B) \equiv \neg A \vee B.$$

Therefore, we have

$$((L \vee (\neg M)) \implies (\neg K)) \equiv (\neg(L \vee (\neg M)) \vee (\neg K)).$$

Then, by De Morgan's law, we get

$$(\neg(L \vee (\neg M)) \vee (\neg K)) \equiv ((\neg L \wedge M) \vee (\neg K)).$$

Thus, we have

$$\neg((\neg L \wedge M) \vee (\neg K)) \wedge (\neg L).$$

Applying De Morgan's law again, we get

$$\neg((\neg L \wedge M) \vee (\neg K)) \wedge (\neg L) \equiv (\neg(\neg L \wedge M) \wedge \neg(\neg K)) \wedge (\neg L).$$

Applying De Morgan's law once more, we have

$$(\neg(\neg L \wedge M) \wedge \neg(\neg K)) \wedge (\neg L) \equiv ((L \vee \neg M) \wedge K) \wedge (\neg L).$$

By the associative law, this is equivalent to

$$(L \vee \neg M) \wedge K \wedge (\neg L).$$

As a final step, we construct a truth table:

| $L$ | $M$ | $K$ | $\neg L$ | $L \vee \neg M$ | $(L \vee \neg M) \wedge K$ | $(L \vee \neg M) \wedge K \wedge (\neg L)$ |
|-----|-----|-----|----------|-----------------|----------------------------|--------------------------------------------|
| T   | T   | T   | F        | T               | T                          | F                                          |
| T   | T   | F   | F        | T               | F                          | F                                          |
| T   | F   | T   | F        | T               | T                          | F                                          |
| T   | F   | F   | F        | T               | F                          | F                                          |
| F   | T   | T   | T        | F               | F                          | F                                          |
| F   | T   | F   | T        | F               | F                          | F                                          |
| F   | F   | T   | T        | T               | T                          | T                                          |
| F   | F   | F   | T        | T               | F                          | F                                          |

Therefore, there is only one possible combination of truth values for  $K$ ,  $L$ , and  $M$ .

Namely,

- $L$  is false,
- $M$  is false,
- and  $K$  is true.

■

## 1.2 (b)

**Problem 2.** Let  $G$ ,  $H$ , and  $J$  be statement variables. Use a truth table to prove that

$$H \wedge (G \iff J) \equiv (G \wedge (H \wedge J)) \vee \neg(H \implies (G \vee J)).$$

**Solution.** Here is the truth table for  $H \wedge (G \iff J)$

| $H$ | $G$ | $J$ | $G \iff J$ | $H \wedge (G \iff J)$ |
|-----|-----|-----|------------|-----------------------|
| $T$ | $T$ | $T$ | $T$        | $T$                   |
| $T$ | $T$ | $F$ | $F$        | $F$                   |
| $T$ | $F$ | $T$ | $F$        | $F$                   |
| $T$ | $F$ | $F$ | $T$        | $T$                   |
| $F$ | $T$ | $T$ | $T$        | $F$                   |
| $F$ | $T$ | $F$ | $F$        | $F$                   |
| $F$ | $F$ | $T$ | $F$        | $F$                   |
| $F$ | $F$ | $F$ | $T$        | $F$                   |

Here is the truth table for  $(G \wedge (H \wedge J)) \vee \neg(H \implies (G \vee J))$

| $H$ | $G$ | $J$ | $H \wedge J$ | $G \wedge (H \wedge J)$ |
|-----|-----|-----|--------------|-------------------------|
| $T$ | $T$ | $T$ | $T$          | $T$                     |
| $T$ | $T$ | $F$ | $F$          | $F$                     |
| $T$ | $F$ | $T$ | $F$          | $F$                     |
| $T$ | $F$ | $F$ | $F$          | $F$                     |
| $F$ | $T$ | $T$ | $F$          | $F$                     |
| $F$ | $T$ | $F$ | $F$          | $F$                     |
| $F$ | $F$ | $T$ | $F$          | $F$                     |
| $F$ | $F$ | $F$ | $F$          | $F$                     |

| $H$ | $G$ | $J$ | $G \vee J$ | $H \implies (G \vee J)$ | $\neg(H \implies (G \vee J))$ |
|-----|-----|-----|------------|-------------------------|-------------------------------|
| $T$ | $T$ | $T$ | $T$        | $T$                     | $F$                           |
| $T$ | $T$ | $F$ | $T$        | $T$                     | $F$                           |
| $T$ | $F$ | $T$ | $T$        | $T$                     | $F$                           |
| $T$ | $F$ | $F$ | $F$        | $F$                     | $T$                           |
| $F$ | $T$ | $T$ | $T$        | $T$                     | $F$                           |
| $F$ | $T$ | $F$ | $T$        | $T$                     | $F$                           |
| $F$ | $F$ | $T$ | $T$        | $T$                     | $F$                           |
| $F$ | $F$ | $F$ | $F$        | $T$                     | $F$                           |

| $H$ | $G$ | $J$ | $(G \wedge (H \wedge J)) \vee \neg(H \implies (G \vee J))$ |
|-----|-----|-----|------------------------------------------------------------|
| $T$ | $T$ | $T$ | $T$                                                        |
| $T$ | $T$ | $F$ | $F$                                                        |
| $T$ | $F$ | $T$ | $F$                                                        |
| $T$ | $F$ | $F$ | $T$                                                        |
| $F$ | $T$ | $T$ | $F$                                                        |
| $F$ | $T$ | $F$ | $F$                                                        |
| $F$ | $F$ | $T$ | $F$                                                        |
| $F$ | $F$ | $F$ | $F$                                                        |

Since  $H \wedge (G \iff J)$  and  $(G \wedge (H \wedge J)) \vee \neg(H \implies (G \vee J))$  have the same truth table value. Thus, these two statements are equivalent.

■

## 2 Problem 2

(a):

**Problem 3.** State the hypothesis of  $J(q, r)$

**Solution.**  $q + 2 \neq 0$  or  $r^2 - 1 \neq 0$ . ■

(b)

**Problem 4.** State the conclusion of  $J(q, r)$

**Solution.**

$$(q + 2)(r^2 - 1) \neq 0$$

■

(c)

**Solution.**

$L(q, r) : \text{"If } (q + 2)(r^2 - 1) = 0, \text{ then } q + 2 = 0 \text{ and } r^2 - 1 = 0."$

That is false. ■

(d)

**Solution.**

$K(q, r) : \text{"If } (q + 2)(r^2 - 1) \neq 0, \text{ then } q + 2 \neq 0 \text{ or } r^2 - 1 \neq 0."$

That is true. ■

(e)

**Solution.**

$$\forall r \in \mathbb{R}, \exists q \in \mathbb{R}, [(q + 2 \neq 0) \vee (r^2 - 1 \neq 0) \implies ((q + 2)(r^2 - 1) \neq 0)]$$

This statement is true. ■

(f): This statement is true.

### 3 Problem 3

- (a): This is false.

**Solution.** Suppose, there exists five consecutive integers whose average is equal to the second-largest of those integers. Let  $n$  be an arbitrary integer in  $\mathbb{Z}$ . We set these five integers are

$$n-2, \quad n-1, \quad n, \quad n+1, \quad \text{and} \quad n+2.$$

By the assumption, we have

$$n+1 = \frac{(n-2) + (n-1) + n + (n+1) + (n+2)}{5}.$$

By expanding the numerator, we get

$$n+1 = \frac{n-2+n-1+n+n+1+n+2}{5}.$$

By adding up the numerator, we get

$$n+1 = \frac{5n+0}{5} = \frac{5n}{5}.$$

Simplifying, we get

$$n+1 = n,$$

which is a contradiction to the fact that  $n+1 \neq n$ .

Therefore, the original claim does not hold. ■

- (b) This is true.

**Solution.** Suppose that  $k$  is an arbitrary integer, and  $k^2$  is the first perfect square of the three consecutive perfect squares.

Therefore, the three perfect squares are

$$k^2, \quad (k+1)^2 \quad (k+2)^2.$$

Take average of these three perfect squares, we have

$$avg = \frac{k^2 + (k+1)^2 + (k+2)^2}{3}.$$

By expanding each term and adding them together, we get

$$avg = \frac{k^2 + k^2 + 2k + 1 + k^2 + 4k + 4}{3} = \frac{3k^2 + 6k + 5}{3}.$$

Then, we can split  $3k^2 + 6k + 5$  into  $3k^2 + 6k + 3 + 2$ , we get

$$avg = \frac{3k^2 + 6k + 3 + 2}{3} = \frac{3k^2 + 6k + 3}{3} + \frac{2}{3} = k^2 + 2k + 1 + \frac{2}{3}.$$

So there exists  $c = k^2 + 2k + 1$  in  $\mathbb{Z}$  such that

$$avg = c + \frac{2}{3}.$$

Thus, we proved that the average of three consecutive perfect squares is always a non-integer. ■

## 4 Q4

- (a)

**Problem 5.** Prove the following statement:

$$\forall x \in \mathbb{R}, \quad 2|5x + 15| - 4|3x - 18| \leq 90.$$

**Solution.** Let us take a look at the absolute value first.

First, we find under what condition,  $5x + 15 \geq 0$ .

By subtracting 15 to both sides and dividing 5, we get

$$5x \geq -15$$

$$x \geq -3$$

Therefore, when  $x \geq -3$ ,  $5x + 15 \geq 0$ .

When  $x < -3$ ,  $5x + 15 < 0$ .

Next, we find under which condition,  $3x - 18 \geq 0$ .

By adding 18 to both sides and dividing 3, we get

$$3x \geq 18$$

$$x \geq 6$$

Therefore, when  $x \geq 6$ ,  $3x - 18 \geq 0$ .

When  $x < 6$ ,  $3x - 18 < 0$ .

There are three cases to consider:

$$- \quad x < -3$$

Under this condition, both  $5x + 15$  and  $3x - 18$  are less than 0. So we have

$$2[-(5x + 15)] - 4[-(3x - 18)].$$

Expanding the parentheses, we have

$$2[-5x - 15] - 4[-3x + 18].$$

Expanding the brackets, we have

$$-10x - 30 + 12x - 72.$$

Adding all the terms together, we get

$$2x - 102$$

Since we assume  $x < -3$ , we get

$$2x - 102 < 2(-3) - 102 = -6 - 102 = -108 \leq 90.$$

Thus, this statement holds true.

$$- \quad -3 \leq x < 6$$

Under this condition,  $5x + 15$  is positive,  $3x - 18$  is negative.

So we have

$$2|5x + 15| - 4|3x - 18| = 2(5x + 15) - 4[-(3x - 18)]$$

Expanding the bracks, we have

$$10x + 30 - 4[-3x + 18] = 10x - 30 + 12x - 72.$$

Simplifying, we get

$$22x - 42$$

Since  $x < 6$  we get

$$22x - 42 < 22(6) - 42 = 90.$$

Therefore, the inequility holds for  $-3 \leq x < 6$ .

– Case 3:  $x \geq 6$ .

Under this condition, both  $5c + 15$  and  $3x - 18$  are positive.

So we have

$$2|5x + 15| - 4|3x - 18| = 2(5x + 15) - 4(3x - 18)$$

Expanding the bracks, we have

$$10x + 30 - 12x + 72$$

Simplifying, we get

$$-2x + 102$$

Since  $x \geq 6$ , we have

$$-2x + 102 \leq -2(6) + 102 = 90.$$

Since this inequility holds for  $x < -3$ ,  $-3 \leq x < 6$ , and  $x \geq 6$ . Therefore, it holds for all  $x$  in real number set.

■

## 5 Q5

**Problem 6.** Prove the following statement:

$$\forall p, q, r \in \mathbb{R}, \quad 2(p^2 + 13r^2) + q(5q + 4p - 10r) \geq 2pr.$$

**Solution.** Suppose  $p, q, r$  are all arbitrary real numbers in  $\mathbb{R}$ .

According to the real numbers property, we have

$$(p + r)^2 \geq 0 \quad (p + 2q)^2 \geq 0 \quad (q - 5r)^2 \geq 0,$$

Adding all the inequilities together, we get

$$(p + r)^2 + (p + 2q)^2 + (q - 5r)^2 \geq 0.$$

Following from the square of binomial,

$$(p^2 + 2pr + r^2) + (p^2 + 4pq + 4q^2) + (q^2 - 10qr + 25r^2) \geq 0.$$

Expanding the parenthesis, we have

$$p^2 + 2pr + r^2 + p^2 + 4pq + 4q^2 + q^2 - 10qr + 25r^2 \geq 0.$$

Simplifying the inequility, this yields:

$$2p^2 + 26r^2 + 5q^2 + 4pq - 10qr + 2pr \geq 0.$$

Substracting  $2pr$  for both sides, we have

$$2p^2 + 26r^2 + 5q^2 + 4pq - 10qr \geq -2pr.$$

Factoring 2 out of  $2p^2 + 26r^2$  and  $q$  out of  $5q^2 + 4pq - 10qr$ , we get

$$2(p^2 + 13r^2) + q(5q + 4p - 10r) \geq 2pr.$$

Therefore, the original inequility is true. ■

**6   Q6**

- (a) Open Sentence 1:  $P(x)$
- Open Sentence 2:  $T(x)$
- Open Sentence 3:  $S(x)$

- (b) Open Sentence 4:  $U(x)$
- Open Sentence 5:  $M(x)$
- Open Sentence 6:  $Q(x)$

(c)

$$\exists x \in \mathbb{R}, ((x-4)^2(x-6) \neq 0) \wedge (x^2 + 10x + 24 \neq 0) \wedge (x^2 - 10x + 24 \neq 0)$$

(d) Let's say. Pick 0 to be  $x \in \mathbb{R}$ , we have

$$(x - 4)^2(x - 6) = (0 - 4)^2(0 - 6) = (-4)^2 \cdot (-6) = 16 \cdot -6 \neq 0.$$

and

$$x^2 + 10x + 24 = 0^2 + 10(0) + 24 = 24 \neq 0$$

and

$$x^2 - 10x + 24 = 0^2 - 10(0) + 24 = 24 \neq 0$$

Therefore, there exists  $x$  in  $\mathbb{R}$  such that  $(x - 4)^2(x - 6) \neq 0$ ,  $x^2 + 10x + 24 \neq 0$ , and  $x^2 - 10x + 24 \neq 0$ .